Engineering Projects Summary

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Project I: HYBRID SPINDLE LINER

PROBLEM

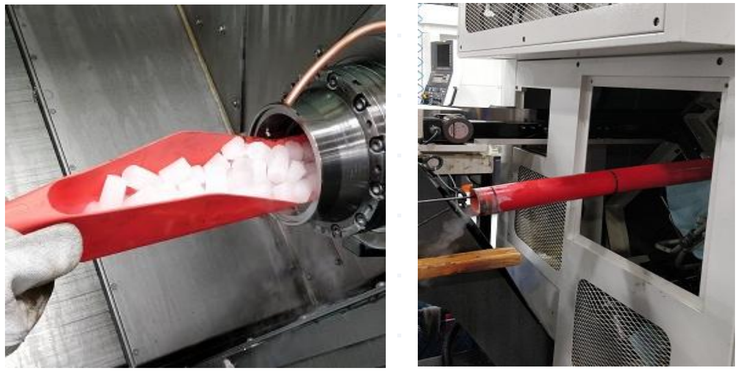
- Polyurethane spindle liners are used in CNC lathes to reduce vibration and improve machining finishes. However, they absorb coolant over time, causing them to **expand and seize** in the machine.
- Stuck spindle liners require **dry ice and lengthy labor** to extract, causing **significant machine downtime** during part changeovers.

METHOD

- Engineered a **hybrid** spindle liner solution consisting of a permanent **metal liner** with a removable **polyurethane** section.
- Designed a bayonet mechanism for **rapid installation and removal** on **Creo.**
- Developed a custom tool to aid installation and removal from the sub-spindle chuck side, eliminating the need for rear panel disassembly.

RESULT

- Produced a functional prototype that may reduce spindle liner changeover times from **over 30 minutes to under 5 minutes.**
- Opened a potential for machine coolant flow improvement by introducing a customizable internal bore within the polyurethane liner.

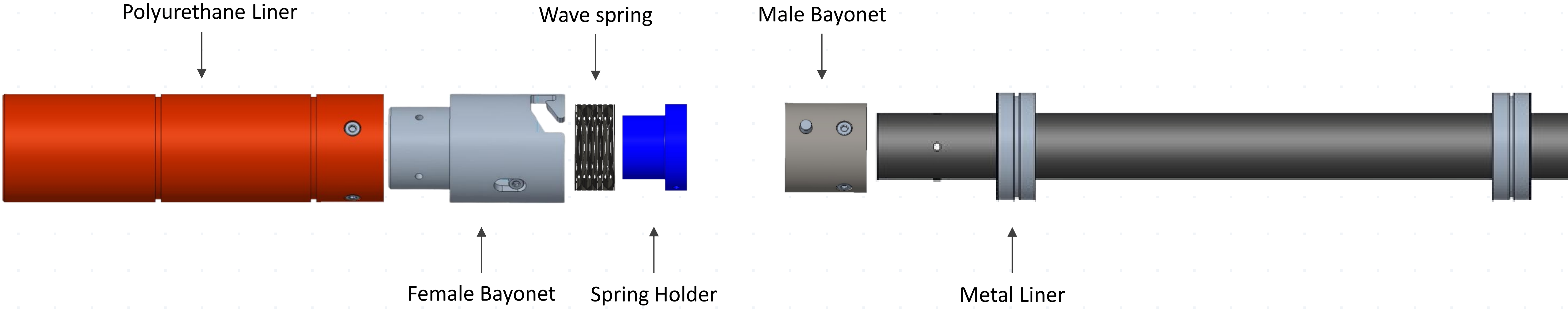


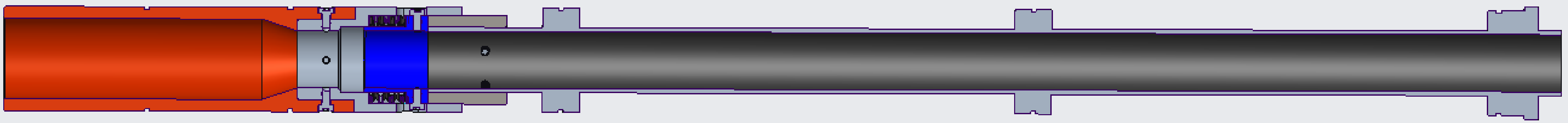
Original removal procedure using dry ice

3D printed prototypes

Custom installation / removal tool

Project I: HYBRID SPINDLE LINER CAD OVERVIEW

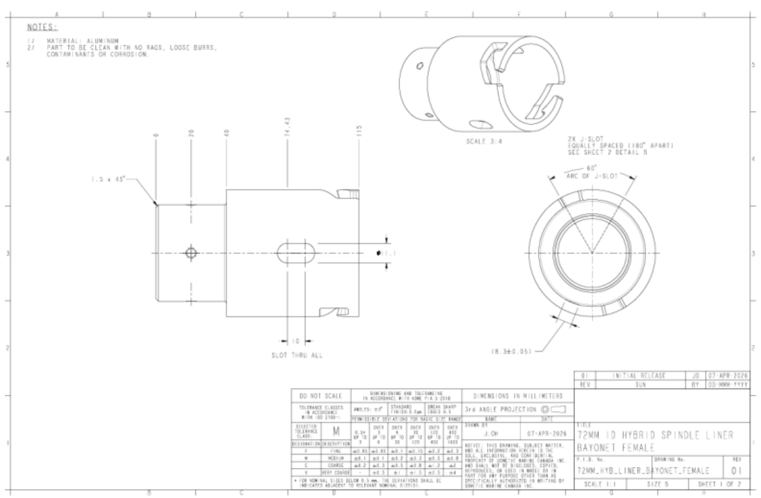


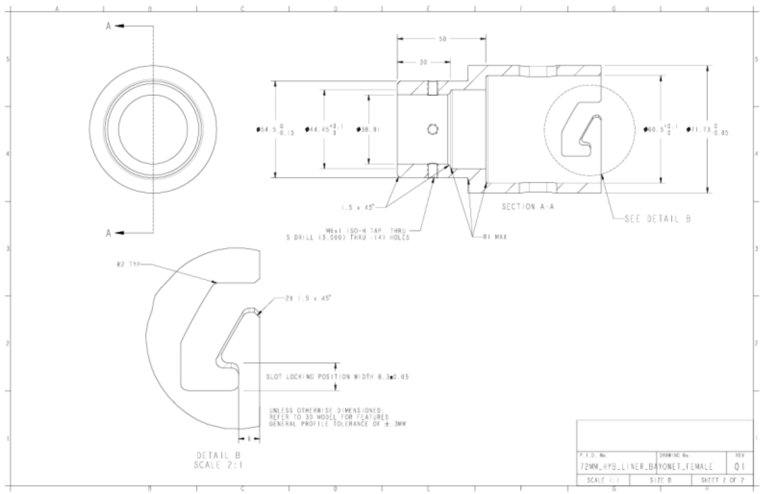


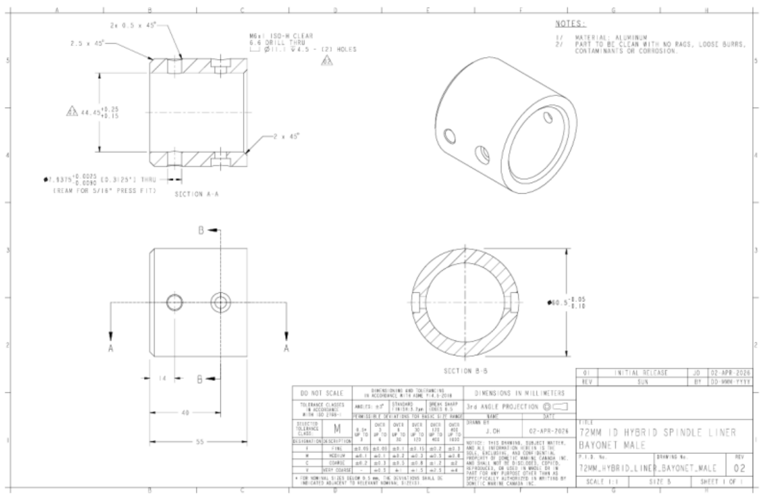
Section view

Project I: HYBRID SPINDLE LINER CONT.

Functional Prototype Assembly





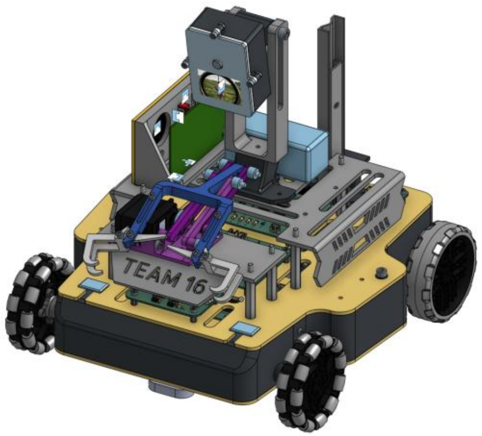


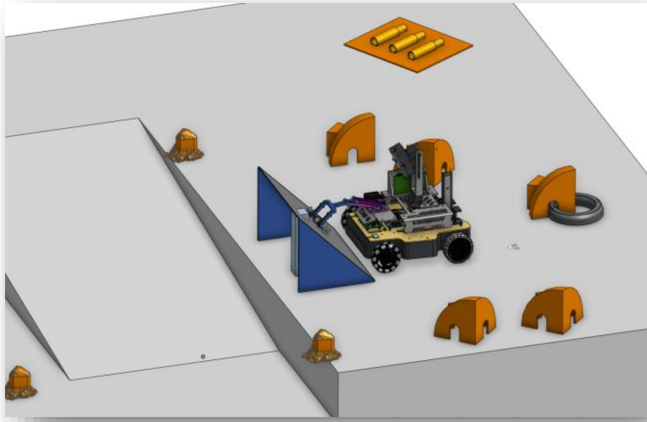
Interface part drawings

Project II: MARS HABITAT ROBOT

PROJECT OVERVIEW

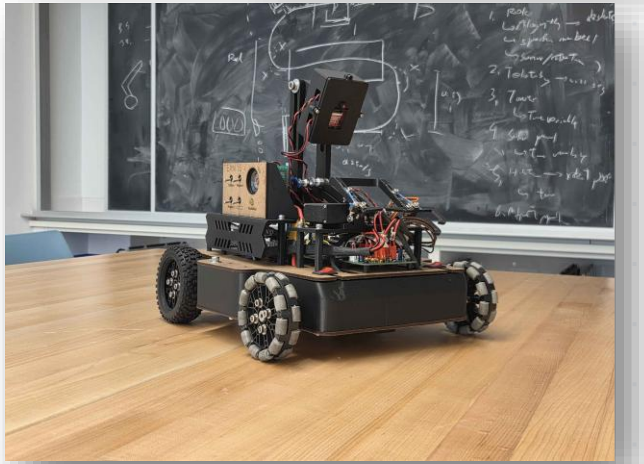
- Tasked with building an **autonomous robot from scratch** in a dedicated 6-week course “ENPH 253 Robot Summer” to navigate a simulated Mars construction site. Missions include independently assembling habitat modules, uncovering model solar panels, and detection of target figures.





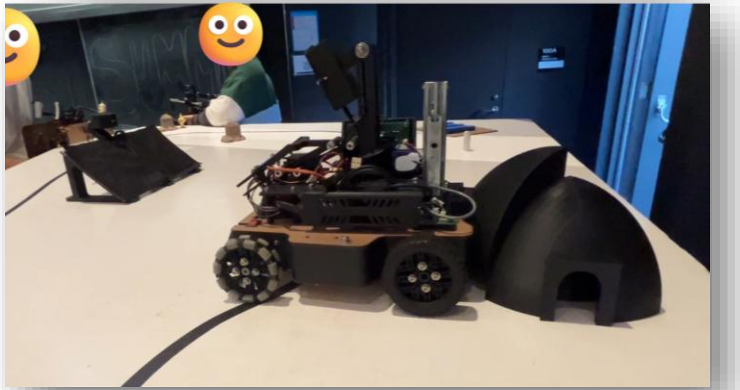
DESIGN

- Modeled the primary chassis, drivetrain, and several electromechanical assemblies using **Onshape and Solidworks.**
- Rapidly prototyped and iterated custom physical components utilizing **3D printing, laser cutting, waterjetting, and traditional machining.**
- Packaged and integrated a sensor suite including **IR sensors, an IMU, custom PCBs, and an ESP32 CAM** module within tight mechanical constraints.



OUTCOME

- Successfully executed autonomous navigation, habitat assembly, and object detection tasks, **scoring 6 points** in the final competition against a **historical mean score of 0.**
- Achieved robust mechanical reliability and high build quality, resulting in consistent performance across multiple competition runs.



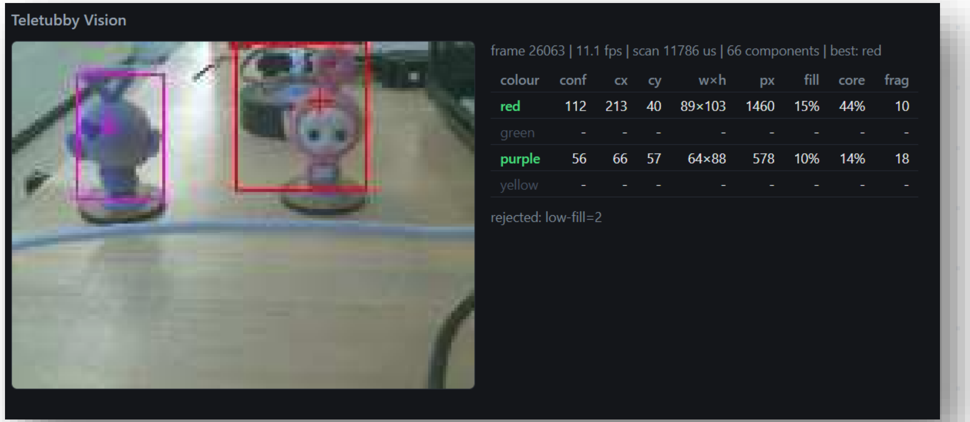
Project II: MARS HABITAT ROBOT CONT.

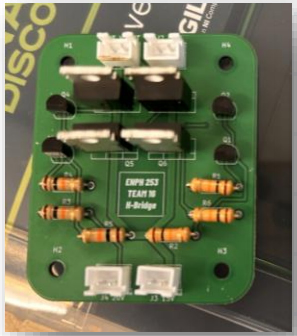
ELECTROMECHANICAL MODULES

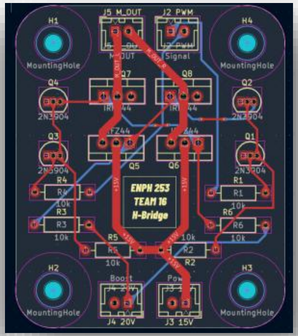
- Fabricated a **user input board** featuring toggle switches for calibration & program toggle, with a LED and a speaker to signal target detection.
- Soldered an **ESP32 CAM** and decoupling capacitors onto a **protoboard**, and designed an enclosure to protect the module while maintaining wired TX/RX communication and program flashing accessible.

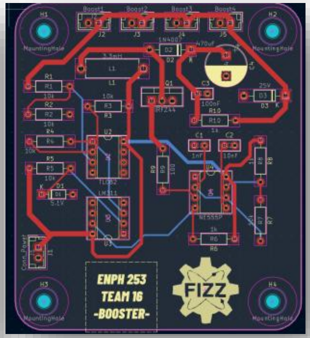
TARGET FIGURE (TELETUBBY) DETECTION

- Implemented **blob detection** using an ESP32 CAM. Maximized accuracy across variable lighting using camera exposure calibration and a **Wi-Fi-hosted HSV tuning suite**. Minimized false positives by calculating a multi-factor **confidence score** (evaluating HSV similarity, blob fill percentage, and aspect ratio).









CUSTOM PCB DESIGN

- Designed a custom **motor driver H-bridge** and a **15V to 20V booster** PCB using **KiCad**; the booster was required for H-bridge functionality under the constraint of exclusively using **NPN MOSFETs**.

Project III: CLAMPING SHAFT COUPLER

PROJECT OVERVIEW

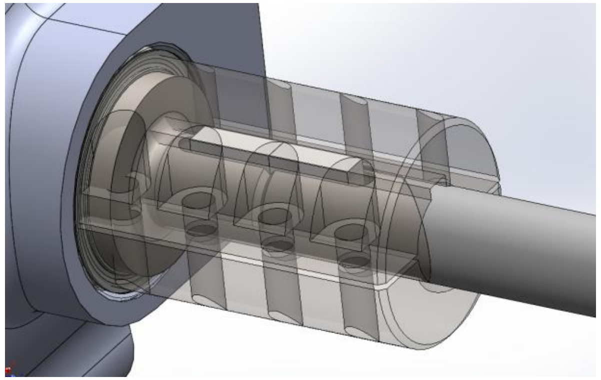
- Designed a custom mechanical interface connecting the reduction **gearbox to the propeller shaft** for the **UBC Baja SAE 4WD vehicle**. Main focus on **weight optimization**, **concentricity**, and **DFM for in-house manufacturing**.

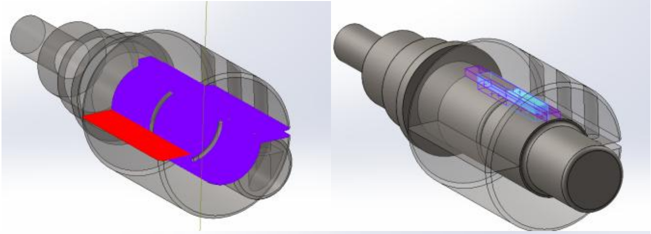
DESIGN

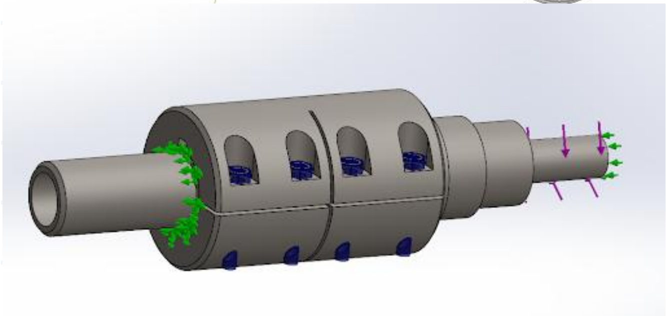
- Modeled the **complete coupling assembly in AISI 4130 / 4340** using **SolidWorks**, integrating the gearbox output, clamping coupler, and a propeller shaft adapter.
- Conducted **assembly FEA** to validate the strength of components (Including keys and fasteners) while actively optimizing for weight.
- Produced a visual manufacturing instruction and engineering drawings to ensure smooth system integration.

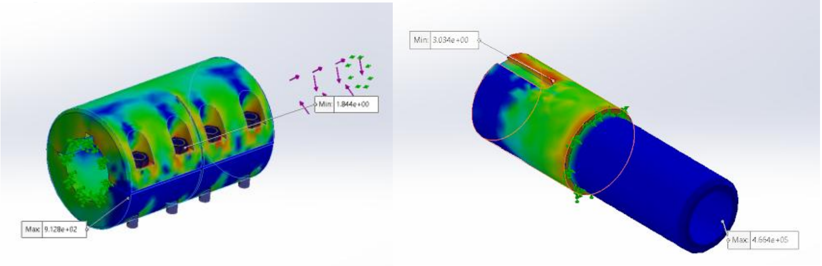
OUTCOME

- Produced a custom coupler design that met our packaging constraints while achieving a **~40% cost reduction** compared to the off the shelf alternatives.
- Successfully powered the 4WD system throughout the **2026 Oregon Baja SAE competition** despite facing initial misalignment issues.





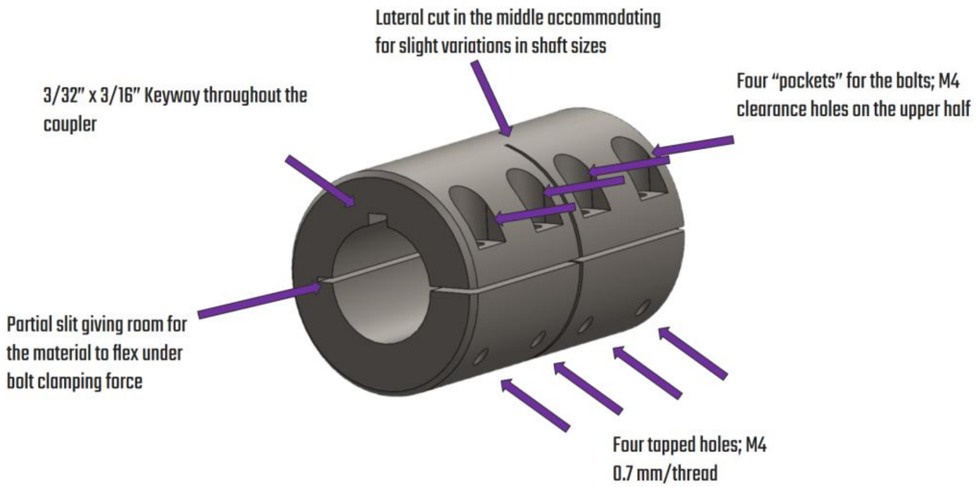




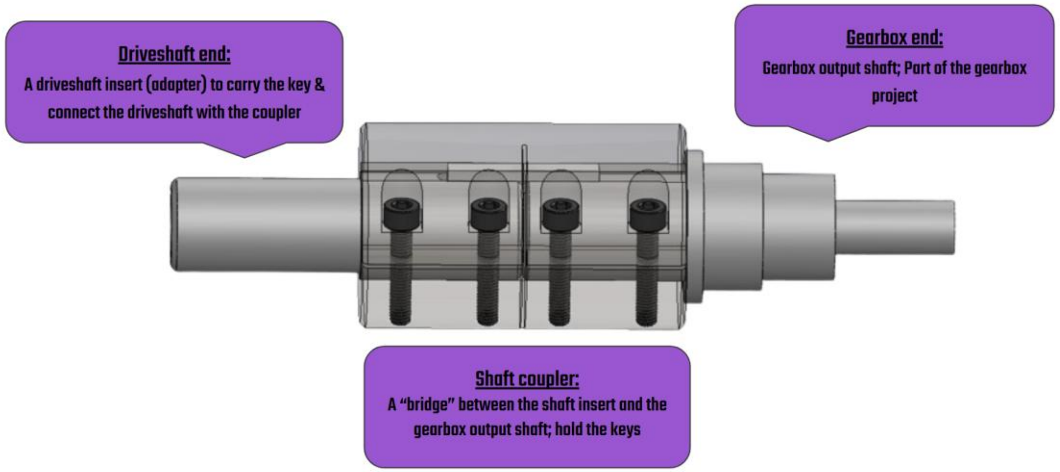
FEA set up and results

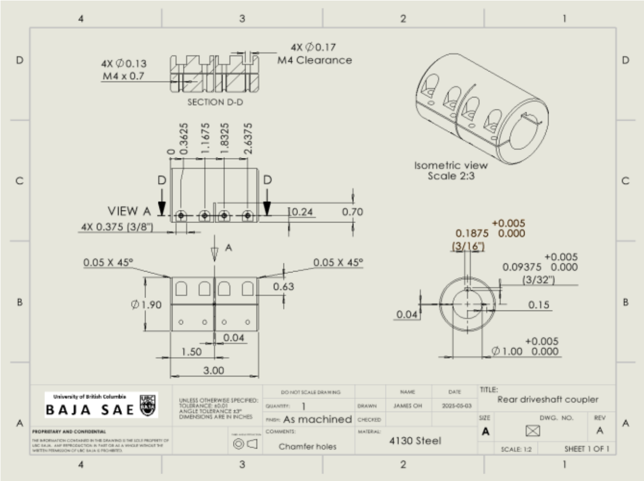
Project III: CLAMPING SHAFT COUPLER CONT.

Coupler overview

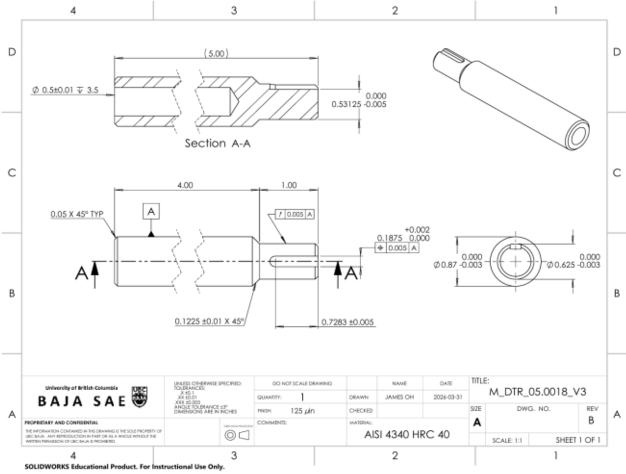


Coupler layout



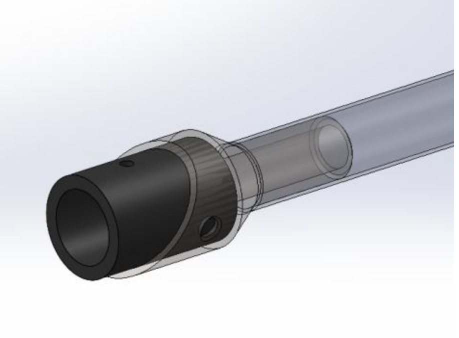


Clamping Coupler Drawing



Propeller Shaft Adapter Drawing

Project IV: SHAFT SPLINE ADAPTER



PROJECT OVERVIEW

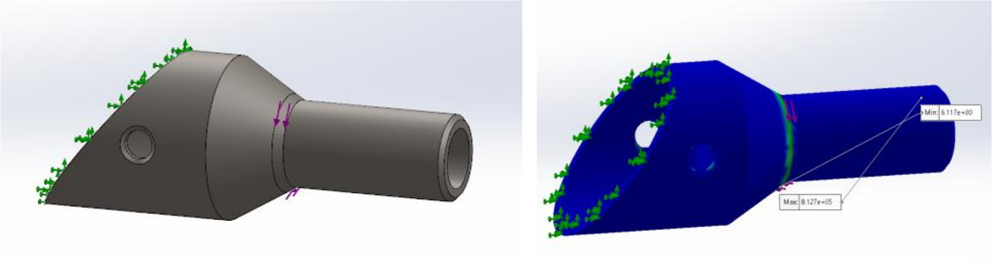
- A custom adapter designed for the **UBC BAJA's 4WD** system connecting the **propeller shaft to the front limited slip differential**. Prioritized **weight optimization, DFM for in-house manufacturing** and **serviceability.**

IMPLEMENTATION

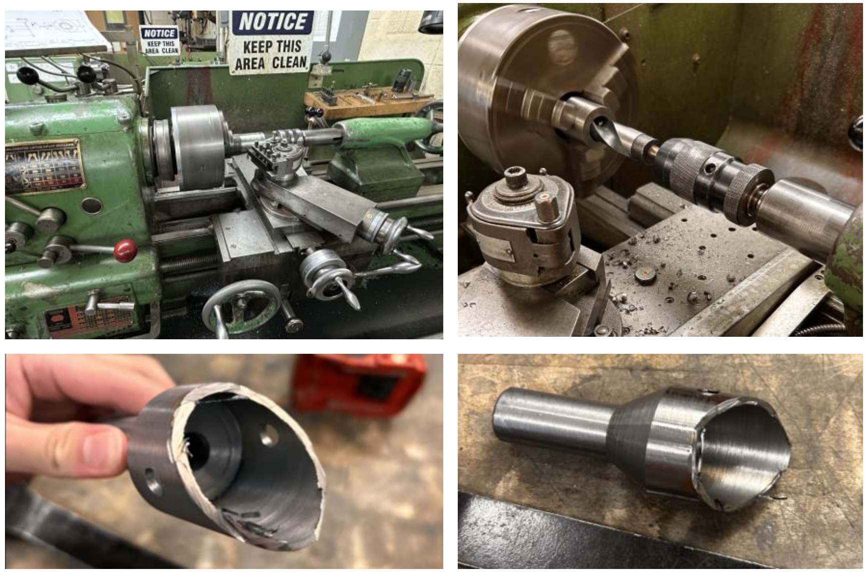
- Designed the connection interface to integrate a female slip spline (Obtained by modifying a Polaris Yoke) fitting for the off-the-shelf (OTS) differential.
- Modeled the primary adapter **in AISI 4340** using **SolidWorks**, integrating the propeller shaft, the adapter, slip spline and the differential.
- Conducted **FEA** to validate the strength of the adapter under torsional loads while meeting dimensional constraints set by the spline and propeller shaft.

OUTCOME

- Independently fabricated the coupler in-house entirely, involving **10+ hours of lathing / milling operations.**
- Successfully powered the 4WD system throughout the **2026 Oregon Baja SAE competition** with zero mechanical failures.

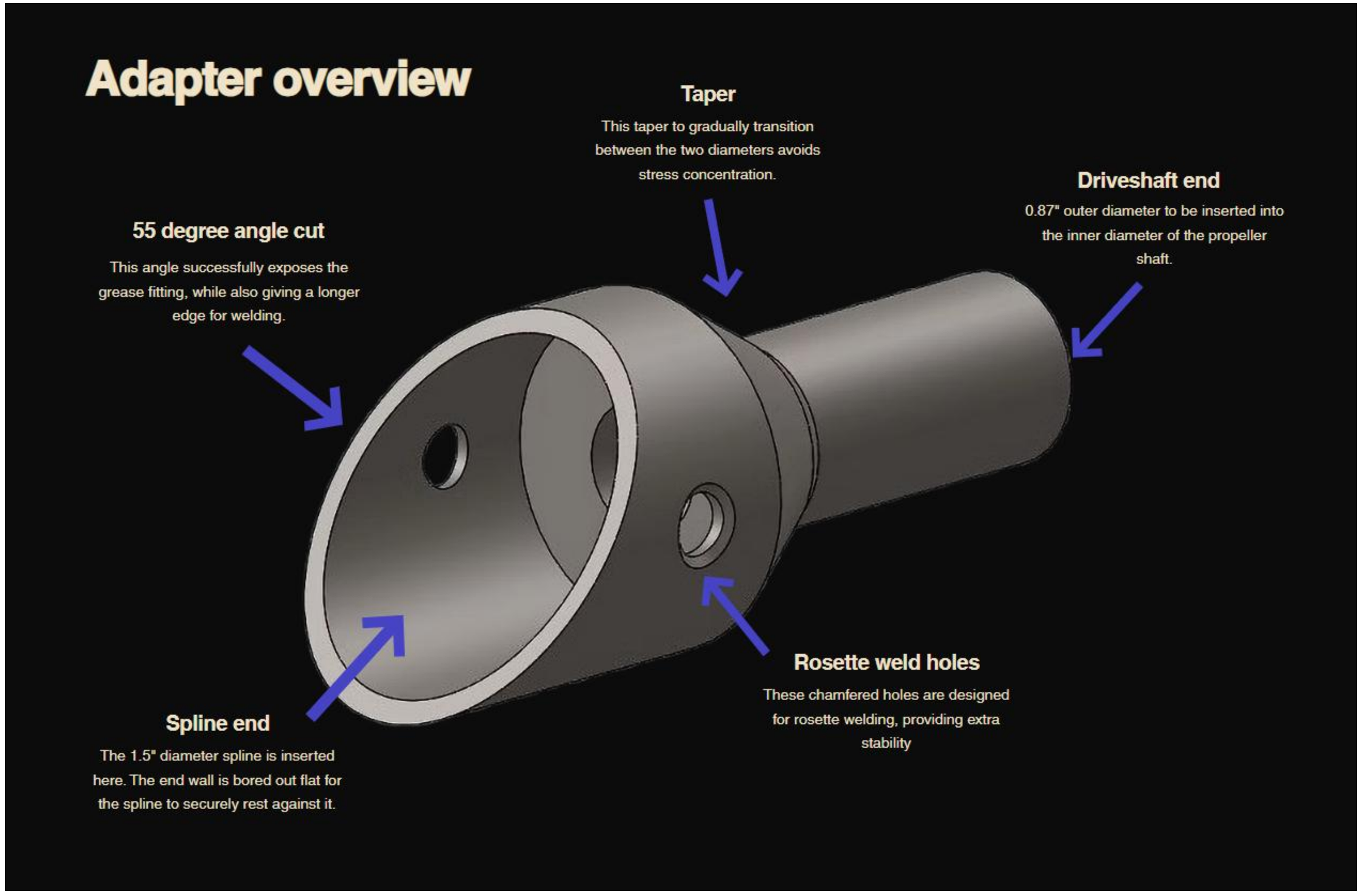


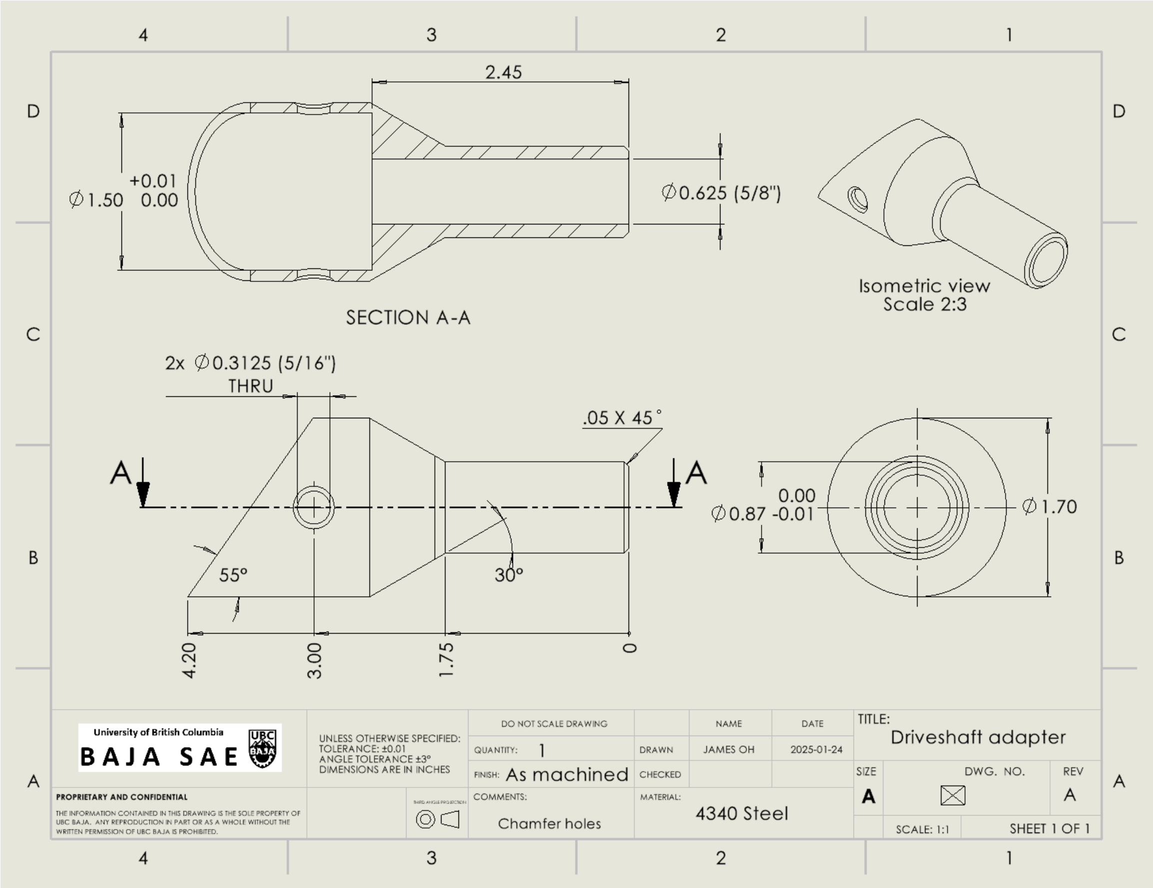
FEA set up and results



Manufacturing (Lathing & Angle Grinding)

Project IV: SHAFT SPLINE ADAPTER CONT.

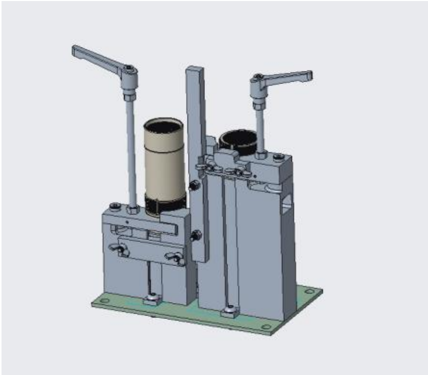


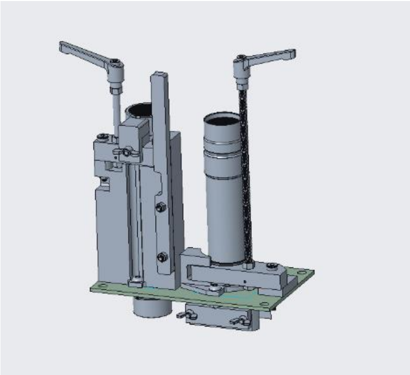


Project V: PRE-NITRIDING PART WASHING JIG

PROBLEM

- Linear actuator rotors require complete removal of grease and machining debris prior to nitriding to ensure even nitrogen absorption.
- The existing ultrasonic washing process required operators to manually hold parts against rotating brushes, creating severe ergonomic **risk for repetitive stress injury** and resulting in **inconsistent internal thread cleaning.**

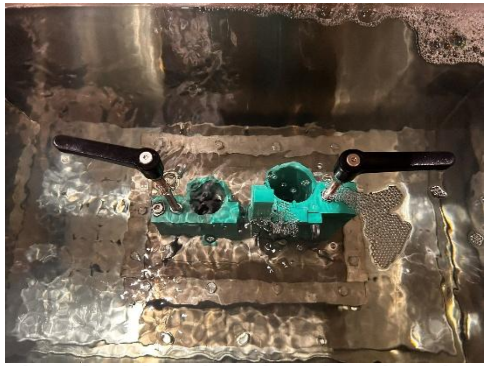




CAD Assemblies (For two different rotor parts)

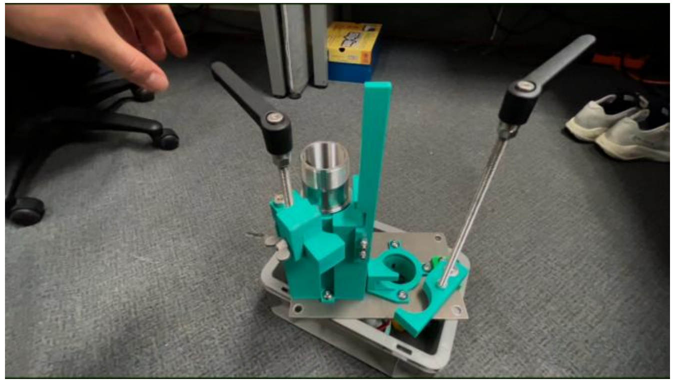
METHOD

- Designed a **rapid locking 3D-printed fixture** to securely and intuitively hold the rotors during the ultrasonic cleaning cycle.
- Conducted tests across various 3D print filaments such as **ABS, ASA, PET-CF** to evaluate thermal, chemical and mechanical resistance in heated liquid bath, ultimately selecting **ASA** for durability and cost effectiveness.



RESULT

- Deployed the fixtures to production for a **under $350 of total budget**, **eliminating repetitive stress injury risks.**
- Authored a **Standard Operating Procedure (SOP)** to ensure consistency, successfully reducing part repeat cleaning rates.



Let's Connect!

I am currently seeking roles as a mechanical / manufacturing engineering intern ☺

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